QSCP Hamilton Inspection Department

Inspection Report

Site: Hamilton Plant

No.	Status	Date
QSCP-INS-2025-014	Issued	2025-04-07

External Visual Inspection and CUI Windows - ATU-100

Scope and Responsibilities

This report documents the 5-yearly external visual inspection and CUI window examinations for the ATU-100. The scope includes all insulated lines within the CUI temperature range as defined by the site CUI program, ensuring compliance with the requirements for profile RT or insulation windows every 5 years. Jennifer Walsh served as the Inspection Coordinator and API 570 inspector, overseeing the field execution and data verification.

The inspection activities were performed in accordance with the guidelines established in API 570 and API 574. Responsibilities included the assessment of piping integrity, verification of insulation condition, and the execution of thickness measurements at designated CMLs to determine remaining life and next inspection dates based on the larger of long-term and short-term corrosion rates.

Inspection Summary and CUI Program Status

In accordance with QSCP-IP-ATU100 Rev 4, CUI windows were opened on all lines currently due under the Rev 2 program. This cycle addresses the access limitations noted in QSCP-INS-2020-017, where the 2-ST-101 CUI window was not opened due to access constraints. For the current 2025 cycle, 2-ST-101 was successfully completed earlier on 2025-03-04.

The external visual inspection focused on identifying signs of coating failure, moisture ingress, and mechanical damage to the insulation cladding. All insulated lines within the CUI range were subjected to window openings to facilitate direct visual examination of the pipe surface and ultrasonic thickness gauging where required by the revised inspection plan.

Piping Inspection Plan and RBI Intervals

The inspection intervals have been updated per QSCP-IP-ATU100 Rev 4. For Class 1 lines, including sour gas, regenerator overhead, and rich amine, the RBI interval is set at 3 years. Specifically, 4-RA-101 is scheduled for re-measurement in Aug 2025 with a planned replacement during the Aug 2026 turnaround. 6-RA-103-11 is scheduled for re-measurement in Feb 2026.

Class 2 and Class 3 lines follow a 6-year RBI interval under the corporate standard. The 2-RF-102 line is scheduled for re-measurement at a 12-month interval in Nov 2025, with replacement planned thereafter. All thickness measurements for A312 TP316 lines were conducted using the specific stainless-steel velocity setting to ensure accuracy against the retirement thickness t-min.

Integrity Assessment and Compliance

Remaining life calculations were performed by Jennifer Walsh using the site rule where remaining life equals (t - t-min) divided by the corrosion rate. The next thickness reading for each circuit is due at no more than half the remaining life, adhering to the class maxima of 5 years for Class 1 and 10 years for Class 2 and 3 piping.

Findings from this 5-yearly external visual inspection confirm that the ATU-100 piping systems generally conform to the requirements of API 570. Any readings found below the line t-min are to be raised as an NCR per site rules. The data collected during this cycle provides the baseline for the next scheduled examinations as defined in the revised piping inspection plan.

Sign-off	Name	Function	Date	Signature
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2025-04-07	_____